

Demonstrating ReCompFig: Designing Dynamically Reconfigurable Kinematic Devices Using Compliant Mechanisms and Tensioning Cables

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ABSTRACT

From creating input devices to rendering tangible information, the field of HCI is interested in using kinematic mechanisms to create human-computer interfaces. Yet, due to fabrication and design challenges, it is often difficult to create kinematic devices that are compact and have multiple reconfigurable motional degrees of freedom (DOFs) depending on the interaction scenarios. In this work, we combine compliant mechanisms (CMs) with tensioning cables to create dynamically reconfigurable kinematic mechanisms. The devices' kinematics (DOFs) is enabled and determined by the layout of bendable rods. The additional cables function as on-demand motion constraints that can dynamically lock or unlock the mechanism's DOFs as they are tightened or loosened. We provide algorithms and a design tool prototype to help users design such kinematic devices. We also demonstrate various HCI use cases including a kinematic haptic display, a haptic proxy, and a multimodal input device.

CCS CONCEPTS

• CCS; • Human-centered computing; • Human computer interaction (HCI); • Interaction devices; • Haptic devices;

KEYWORDS

Haptic proxies, design tool, kinematic devices, wearables, compliant mechanism

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1 INTRODUCTION

In HCI, we are interested in creating devices that afford these kinematic interactions to support a more natural, intuitive, and enriched interaction experience. Along this line of effort, prior works had explored creating haptic proxies for virtual reality [5], displaying material properties and enabling tangible interactions [1], and designing gadgets and toys [3, 4]. However, as we push against the boundary of kinematic device design, creating multimodal (i.e., having more than one kinematic mode) and reconfigurable kinematic devices remains a challenge. The footprints of the mechanical components also impose a size limit, making it difficult to create miniaturized and compact interactive devices. Moreover, adding reconfigurability to the devices also requires latching or locking mechanisms, further exacerbating the design and fabrication complexity. As a result, existing interactive kinematic devices are often highly specialized, only possess a single kinematic mode, and cannot adapt nor reconfigure to different use scenarios.

We address these challenges by combining compliant mechanisms (CMs) and tensioning cables to create multimodal and reconfigurable kinematic mechanisms (ReCompFig). Compliant mechanisms are prescribed with DOFs and kinematic behaviors by making local part geometries slender and flexible (i.e., flexural elements) [2]. A CM can also be designed to offer multiple DOFs without

increasing its mechanical complexity, making it a viable method to design compact and multimodal kinematic devices. However, conventional CMs cannot be reconfigured once fabricated. Therefore, we introduce tensioning cables into CMs as a way to reconfigure (i.e., temporarily lock/unlock individual) DOFs during use as well as add stretchable strain sensors to augment their functionalities as input devices, enabling dynamically reconfigurable kinematic behaviors for HCI uses such as different haptic feedback and physical input modes.

2 DESIGN PRINCIPLES

2.1 Expanding Flexure Design Space Using Cables

ReCompFig comprises three building blocks: rod flexure for rigid support and permanent constraints, tensioning cable for degree of freedom/constraint reconfigurability, and elastic cables for sensing.

2.1.1 Rod flexure. In CMs, a rod flexure typically has an aspect ratio between 1:10 to 1:20. It has buckling strengths well above the expected use-load and negligible tensile extensions in normal situations, meaning they cannot buckle nor extend under normal use scenarios, therefore limiting linear motions along its longitudinal axis (Figure 1A).

2.1.2 Tensioning cables. Unlike rod flexures, cables would buckle when subjected to compressive forces due to their extreme aspect ratio (typically larger than 1:40) and low buckling strength, therefore making them “half rods” in a structural sense, i.e., cables behave like rods under tensile loads but are virtually ineffective under compression (Figure 1B). We leverage cables as an on-demand, reconfigurable flexure element to design CMs. When controlled by a motor, the cable pair can be further shifted in or out of effect when tightened or loosened, respectively, thus enabling reconfiguration for different interaction scenarios without modifying its structure (Figure 1D).

2.1.3 Elastic sensing cables. On the other hand, cables made of elastic materials hold no structural functions in a CM because their elasticity allows them to stretch under tension, thus cannot be used as flexures in any way. However, we can take advantage of this fact and add stretchable sensors (e.g., conductive rubber cord stretch sensors) to detect its deformations without modifying its DOFs (Figure 1C).

2.2 Designing Multimodal Kinematic Mechanisms

The cable-driven reconfiguration can be used to create devices with dynamic and distinct kinematic modes and affordances. First, we can identify their corresponding constraint spaces given the desired DOFs for each kinematic mode. We can then use these constraint spaces to determine the placement of the non-reconfigurable rods – the flexural rods shared by all kinematic modes. At this point, the mechanism should have the combined DOFs of all kinematic modes. Next, tensioning cables are added for each kinematic mode to constrain undesired DOFs, and a control scheme (i.e., a table that matches cable group status with kinematic modes) is generated. Stretchable sensors are also optionally added to the device following

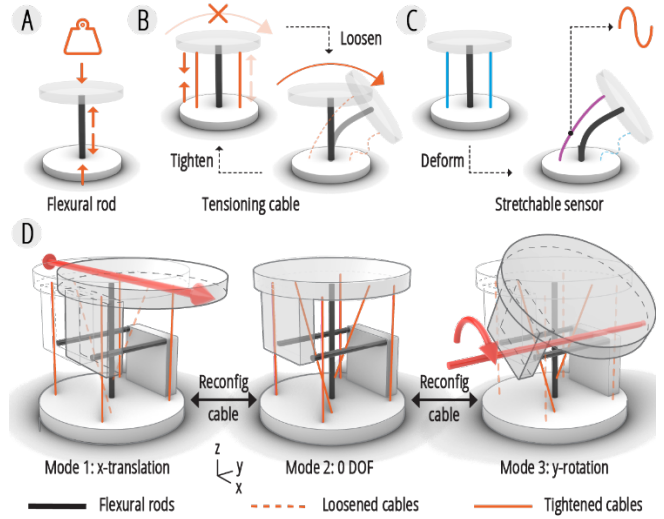


Figure 1: Design principles of reconfigurable, sensing, and multimodal kinematic devices. (A) A flexural rod constrains DOFs by resisting compression and extension, whereas (B) a cable only constrains extension when tightened, and (C) an elastic cable (sensor) imposes no constraint. (D) The tensioning cables can be tightened/loosened to reconfigure the device's degree of freedom.

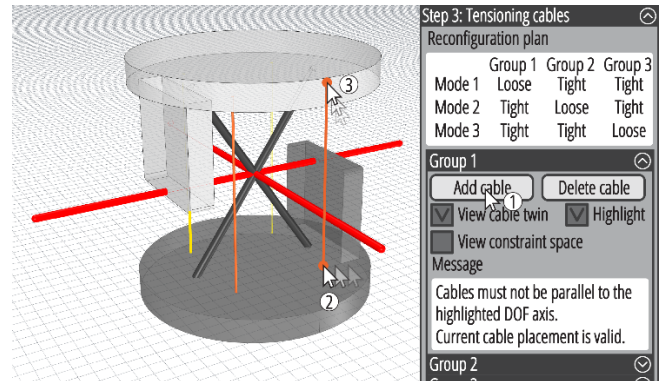


Figure 2: The design tool provides visual and textual prompts to guide users through designing ReCompFig devices

the tensioning cables. Finally, the model is completed by assigning a thickness to the rods and designing guiding tubes for cables and housings for motors. These principles are implemented in a design tool that allows users to import custom geometries and specify desired kinematic modes (Figure 2).

3 APPLICATION EXAMPLES

3.1 Multimodal Input Device

Here, we demonstrate that as a multimodal interface, a ReCompFig input device can support various kinematic interactions at a time (Figure 3). The device can change between three kinematic modes

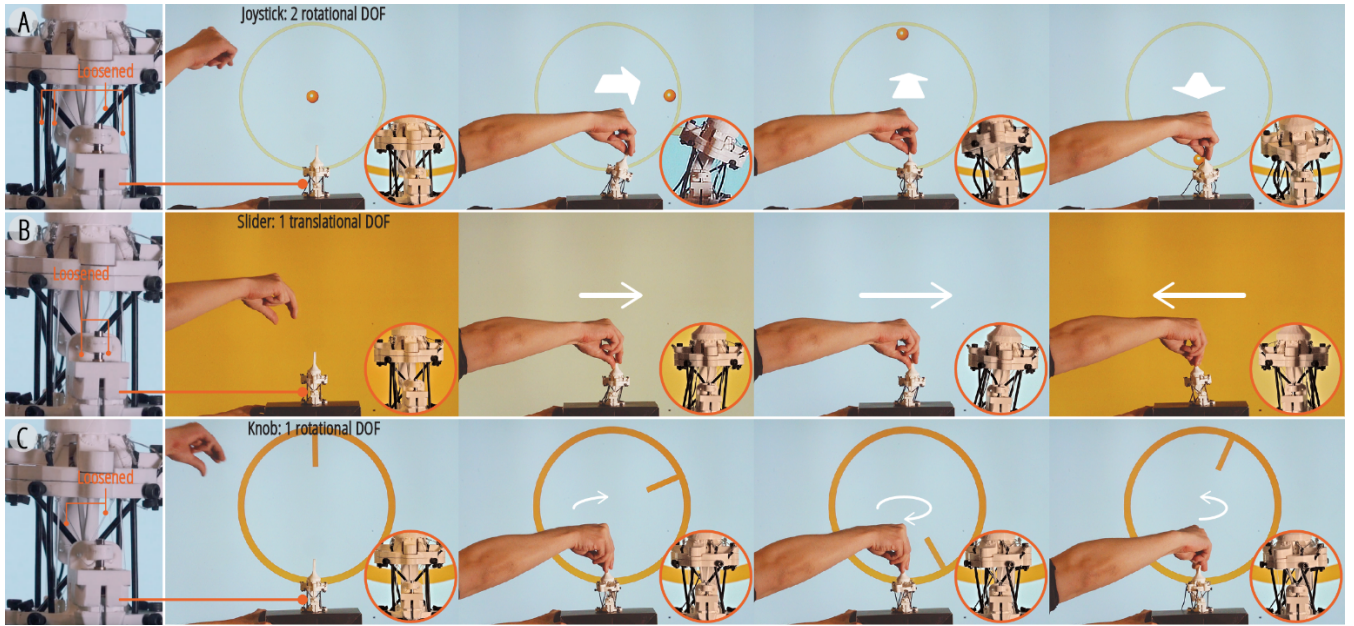


Figure 3: The multimodal input device functions as (A) a joystick to move a ball on the screen in the background, (B) a slider controlling the color on the screen, and (C) a twisting knob to rotate a digital dial. The leftmost images show the cable status under each mode (blue: loosened, orange: tightened), and the insets show zoomed-in views of the device.

on demand, each recreating the haptic experience of a familiar interface (i.e., joystick, slider, twisting knob).

3.2 Kinematic Material Display

In this example, we leverage the compactness and reconfigurability of ReCompFig mechanisms to create a kinematic material display. The display comprises a 4x4 grid of individually addressable kinematic bits (Figure 4A). The modules have a pair of cables to enable/disable their translational and rotational DOFs (Figure 4B).

3.3 Wearable Haptic Proxy

We demonstrate that ReCompFig mechanisms can also be used to create kinematic haptic proxies that simulate the handfeel of holding objects made of different materials (Figure 5A), similar to that of ElaStick [5]. Partial cable tightening simulates unidirectional weight shift along a single axis. This design allows us to simulate the weight shift of holding different objects, including liquids, elastic sticks, and rigid bars (Figure 6A, B, and C, respectively).

4 CONCLUSION

This work introduces tension cables into compliant mechanisms to create multimodal and reconfigurable kinematic mechanisms and devices. In particular, the design tool assists users in designing two aspects of a reconfigurable CM device - its prescribed DOF configurability and the cables used for dynamically switching their DOF modes. The design tool provides procedural and open-ended guidance to assist users in creating mechanisms with the desired kinematic modes and sensing capabilities.

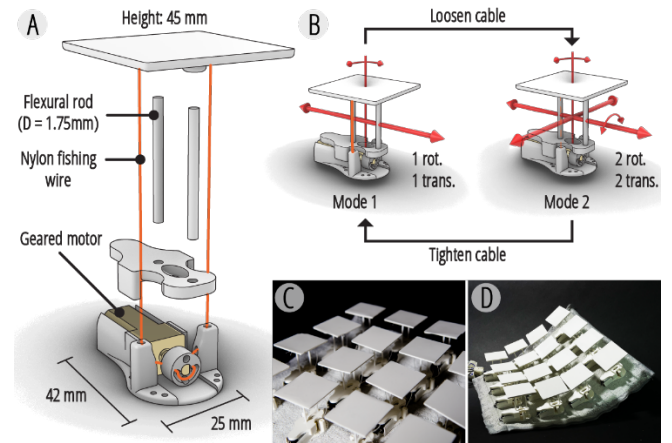


Figure 4: Kinematic material display - (A) unit module design and (B) DOFs before and after cable tightening. (C) Arrays of kinematic modules as an array-like display. (D) The display conforming to a curved surface.

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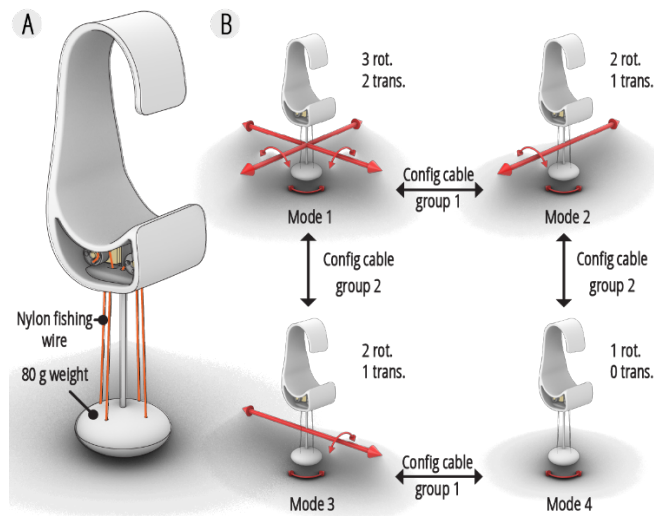


Figure 5: Kinematic haptic proxy - (A) its structure and (B) kinematic modes controlled by a pair of motors.

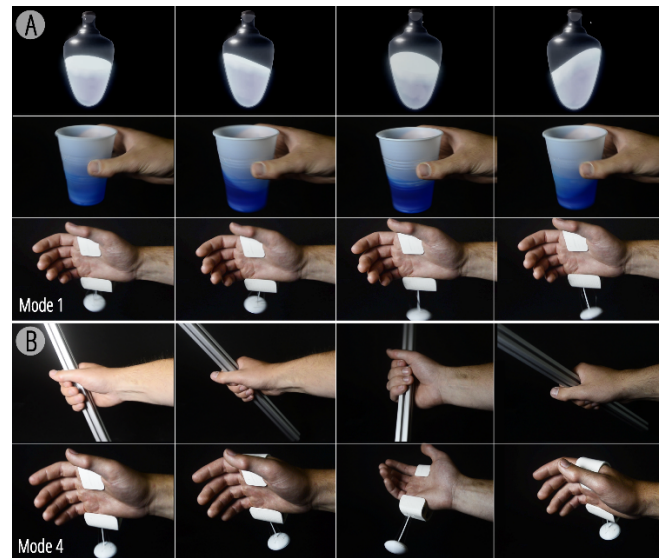


Figure 6: Simulating the haptic feedback of (A) liquids and (B) rigid bars with the kinematic haptic proxy.

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